

CSCE 638 NATURAL LANGUAGE PROCESSING · MODULE 13 · LECTURE NOTES

Claiming Language Understanding Honestly

What a model's performance establishes.

MODULE OUTCOMES

- State what a benchmark result establishes.
- Identify the standard overclaims.
- Write a defensible system claim.
- Place this course relative to CSCE 676 and CSCE 670.
- State what you would need to claim understanding.

1 What a score establishes

A BENCHMARK SCORE IS A STATEMENT ABOUT THAT DATASET, THAT METRIC, AND THAT SPLIT

It establishes that this model, under this decoding configuration, scored this value on these examples by this measure — all four clauses, and none of them optional (Module 09 §4's reporting requirement).

It does not establish generalisation to your data, unless your data happens to be distributed like the benchmark's — **which is an empirical question with a usually negative answer**, and one you can settle cheaply by evaluating on your own held-out set.

And it does not establish the *ability* the benchmark is named after, because **a benchmark may be solvable by an artefact** — which Module 10 §2's shortcut check detects, and which has invalidated several well-known datasets.

Nor does it rule out contamination (Module 10 §4) — **so a high score on a public benchmark is weak evidence**, and **your own held-out data, collected after the training cutoff, is considerably stronger** even if it is smaller.

The claim	The correction
'The model understands X.'	It produces text consistent with understanding X on these inputs — which is a narrower and checkable claim (§3).
'It achieves human-level performance.'	On this dataset, by this metric, against these particular annotators — see the note.

The claim	The correction
'It can reason.'	It produces reasoning-shaped text; check whether the steps are load-bearing (§3's fourth test).
'It knows that P.'	It asserts P here; check whether it asserts $\neg P$ when asked differently — consistency is the test.
'Hallucination is being solved.'	Rates are falling; the mechanism is structural (Module 12 §3), so a rate is not an elimination.
'It scored 90%.'	With which decoding, which prompt, and what variance across prompts? (Module 09, Module 08 §3.)

The second row is the most consequential of these. 'Human-level' invites a generalisation that the measurement does not support — because the human comparison was performed on the same narrow distribution, frequently with annotators working quickly on an unfamiliar task, and the model's failures are distributed quite differently from the humans'. Matching an average on one dataset is not matching a capability. This overclaim does the most damage in public discussion, because the phrase travels and the caveats do not.

2 Claims you can defend

- 'On 2,000 held-out examples from our own data, F1 is 0.84, against a tuned BM25 baseline of 0.71, with a 95% confidence interval of ± 0.02 .' Checkable in every clause, and it names the baseline (Module 10 §2).
- 'Performance on dialectal inputs is 7 points lower than on standard inputs; that estimate is from 300 examples.' Disaggregated, with the sample size stated so a reader can judge the comparison (Module 12 §2).
- 'On questions our corpus cannot answer, the system abstains 62% of the time and fabricates an answer 38% of the time.' A measured failure rate rather than a mitigation claim (Module 11 §4).
- 'Results are with top-p 0.9 at temperature 0.7, averaged over five prompt phrasings; the variance across prompts is 3 points.' Which makes the result reproducible (Module 09 §4, Module 08 §3).
- And: 'we have not evaluated on inputs longer than 500 tokens, or in any language other than English.' The last one is the limitation statement, and it is what distinguishes a system report from a demonstration — which is Project 2's hardest-graded requirement, because stating it requires knowing what you did not test.

3 Understanding

THE HONEST POSITION IS THAT THE QUESTION IS NOT SETTLED AND THE EVIDENCE IS SPECIFIC

'It is just predicting the next token' understates it. Predicting text well requires representations that support generalisation — and probing finds real structure in the activations: syntactic dependency information, factual relations, and even spatial and temporal information. **Dismissing that as 'just statistics' is not an argument.**

'It understands language' overstates it. Performance is brittle to paraphrase, sensitive to formatting, and inconsistent under rephrasing of the same question — which is not the profile of robust understanding, and a system that answers differently depending on the word order has not represented the question.

So the useful move is to stop asking the general question and instead test specific capabilities: is the behaviour consistent under paraphrase? does it transfer to a novel composition of familiar parts? are the stated reasoning steps load-bearing?

Each of those is measurable — and measuring them is considerably more informative than any position on the general question, which is where this course asks you to put the effort. **The general question may not have a decidable answer; the specific ones do.**

- **Paraphrase consistency.** Ask the same question in five different ways — a model that answers differently has not represented the question, it has responded to the surface form.
- **Counterfactual sensitivity.** Change the specific fact the answer depends upon, and check that the answer changes — if it does not, the answer was not derived from the fact.
- **Compositional generalisation.** Test a novel combination of familiar parts — which is where these systems reliably break, and which is Module 01 §1's compositionality row returning as an evaluation.
- **Load-bearing reasoning.** Corrupt an intermediate step in a stated chain of reasoning and see whether the final answer changes — if it does not, the steps were decoration rather than computation. This is the sharpest of the five, and it is cheap.
- **And contamination control:** construct the test set from material postdating the training cutoff, or accept explicitly that you cannot rule contamination out and say so (Module 10 §4). Reasoning-shaped text that does not affect the answer is a presentation, and the test for it takes minutes.

4 The semester, and where this leaves you

Course	What it covers	The shared concern
CSCE 638 (this one)	Language as an input and as an output.	Evaluation that measures the thing you care about rather than the thing that is easy (Module 10).
CSCE 676	Finding structure in data at scale.	Distinguishing a pattern that is real from one that was found by looking long enough.

Course	What it covers	The shared concern
CSCE 670	Satisfying an information need.	Relevance as a human judgement rather than a label — and the proxies for it.

All three are courses about measurement as much as about method — because in all three cases the thing you actually want is a human judgement and the thing you have is a proxy for it: a relevance label, a cluster's coherence, a summary's quality. Which makes the choose-the-metric-from-the-use discipline the semester's organising idea, arriving from CSCE 633 Module 12 and applied three different ways.

WHERE THIS COURSE LEAVES YOU

You can explain the arc from counting to dense representations to recurrence to attention to pretraining, and say what each step bought and what it cost — which is considerably more durable than familiarity with any current architecture, since the arc's logic survives the next architecture.

You can build a system: choose and adapt a pretrained model against real constraints, decode deliberately rather than by default, retrieve when the knowledge is external, and evaluate against a baseline that is genuinely hard to beat.

And you know the characteristic failure of everything in this course — fluent, confident, and wrong — along with why it is structural rather than incidental (Module 12 §3) and which mitigations actually reduce it.

The closing rule is the program's, unchanged across twenty-eight courses: state what you measured, state what you assumed, and never claim more than you established. In this subject it means naming the dataset, the metric, the decoding configuration, and the inputs you did not test — because 'it understands' names none of them, and §1's table is what happens when that substitution goes unchallenged.

Where to see this taught differently

Original material only goes so far — hearing the same idea in another voice is how it sticks. These are free, and they cover this module specifically.

Resource	What it covers for this module
Bender & Koller — Climbing towards NLU (free)	§3's question, argued from the sceptical side — and worth rereading now that you have Module 07's evidence.
Rogers et al. — A Primer in BERTology (free)	§3's probing evidence, surveyed — what has actually been found in the representations, with the caveats.
Turpin et al. — Language Models Don't Always Say What They Think (free)	§3's load-bearing test, applied — stated reasoning that does not determine the answer.

Resource	What it covers for this module
Raji et al. — AI and the Everything in the Whole Wide World Benchmark (free)	§1's argument — why a benchmark cannot establish a general capability, developed properly.

Exercises

Do these in code, not on paper, unless the exercise says otherwise. The module is not finished until they run.

- 1 **Take a published claim about a model** and rewrite it in §1's four-clause form.
- 2 **Find a 'human-level' claim** and identify what the human comparison actually measured.
- 3 **Write the defensible claim** for your own Project 2 system.
- 4 **Write its limitation statement** — the inputs you did not test.
- 5 **Run the paraphrase consistency test** with five phrasings of one question.
- 6 **Run the counterfactual test** on a fact-dependent answer.
- 7 **Construct a novel composition of familiar parts** and test it.
- 8 **Corrupt an intermediate reasoning step** and check whether the answer changes.
- 9 **Revisit Module 01's last exercise** and compare your predictions against what you met.
- 10 **Project 2 is now due.** Submit the task and its decision, the tuned baseline, your system, the metric justification, the twenty-error analysis, and the failure and limitation statement.

Self-check

Answer these in writing, from memory, without looking back at the notes. If you cannot, you have read this module but not yet learned it — that is useful information, not a failure.

- 1 What four clauses does a benchmark score require?
- 2 What does it not establish — give three things?
- 3 Give six overclaims and the correction to each.
- 4 Why is 'human-level' the most consequential?
- 5 Give four defensible claim forms.
- 6 What distinguishes a system report from a demonstration?
- 7 Why is 'just predicting the next token' an understatement?
- 8 Why is 'it understands language' an overstatement?
- 9 Give the five discriminating tests and the sharpest one.
- 10 What do the three Semester 10 courses share?